

Home Credit Default Risk

Model Training Pipeline Report

Generated: March 31, 2026

Model: XGBoost Classifier

Test set ROC-AUC: 0.7793

1. Model Performance Overview

The model was evaluated on a held-out test set comprising 20% of the full training data, stratified by the target variable to preserve the class distribution. The primary evaluation metric is ROC-AUC, which measures the model's ability to rank applicants by default risk regardless of the decision threshold chosen.

Metric	Value
ROC-AUC	0.7793
Test set size	61,503
Positive rate (test set)	0.0807

Interpretation: Good result — in the expected range for this dataset with the full supplementary feature set.

2. ROC Curve

The ROC curve plots the true positive rate against the false positive rate at every possible classification threshold. A perfect model would reach the top-left corner. The diagonal line represents a random classifier. The area under this curve (AUC) is the primary model metric.

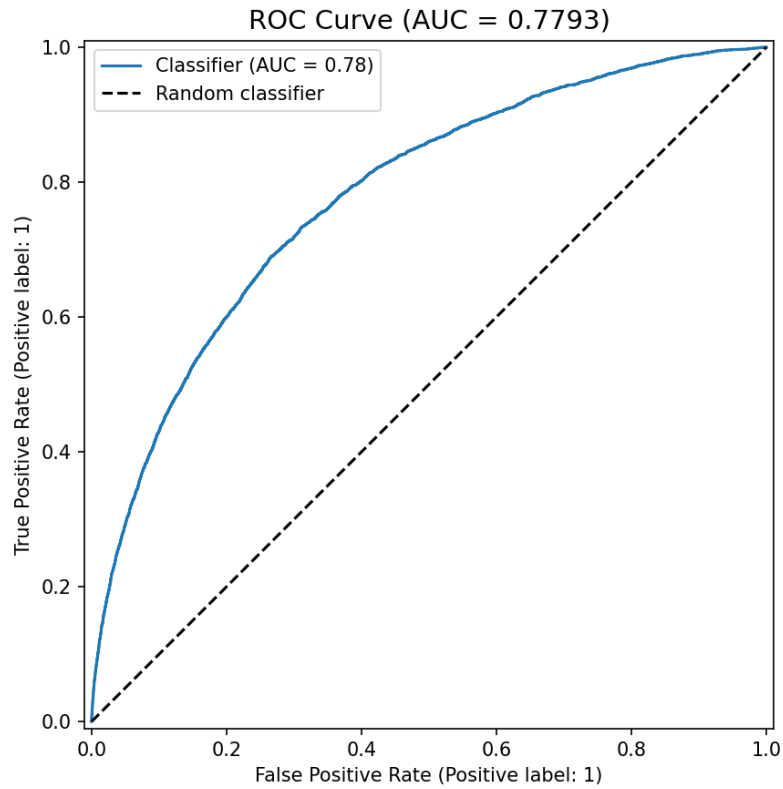


Figure 1. ROC Curve — AUC = 0.7793

3. Classification Report

The classification report shows precision, recall, and F1 score for each class at the default decision threshold of 0.5. Given the class imbalance (approximately 92% repays, 8% defaults), these metrics should be interpreted with caution — the model is optimized for ranking (AUC) rather than for any specific threshold. Precision for the Defaults class answers: of all applicants the model flagged as likely to default, what fraction actually did? Recall answers: of all applicants who actually defaulted, what fraction did the model catch?

Class	Precision	Recall	F1 Score	Support
Repay	0.9611	0.7648	0.8518	56538
Defaults	0.1947	0.6475	0.2994	4965
Weighted avg	0.8992	0.7553	0.8072	61503

4. Confusion Matrix

The confusion matrix shows the count of correct and incorrect predictions at the default threshold of 0.5. True negatives (top-left) are applicants correctly predicted to repay. True positives (bottom-right) are applicants correctly predicted to default. False positives (top-right) are applicants incorrectly flagged as defaulters. False negatives (bottom-left) are defaulters the model missed — in credit risk this is typically the more costly error.

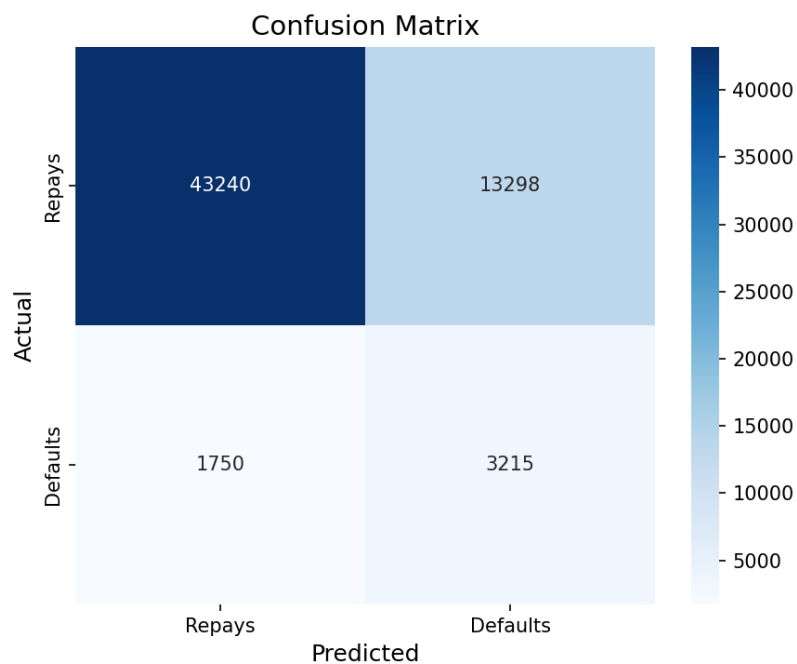


Figure 2. Confusion matrix at default threshold of 0.5.

5. SHAP Feature Importance

SHAP (SHapley Additive exPlanations) values measure each feature's contribution to individual predictions. Unlike XGBoost's native feature importance, SHAP accounts for feature interactions and provides directional information — showing not just which features matter, but whether high or low values of each feature push predictions toward default or repayment.

5.1 Beeswarm Plot

Each dot represents one applicant. Horizontal position shows the SHAP value — how far that feature pushed the prediction away from the baseline. Color represents the feature value: red indicates a high feature value, blue indicates a low feature value. Features are ranked by mean absolute SHAP value with the most important at the top.

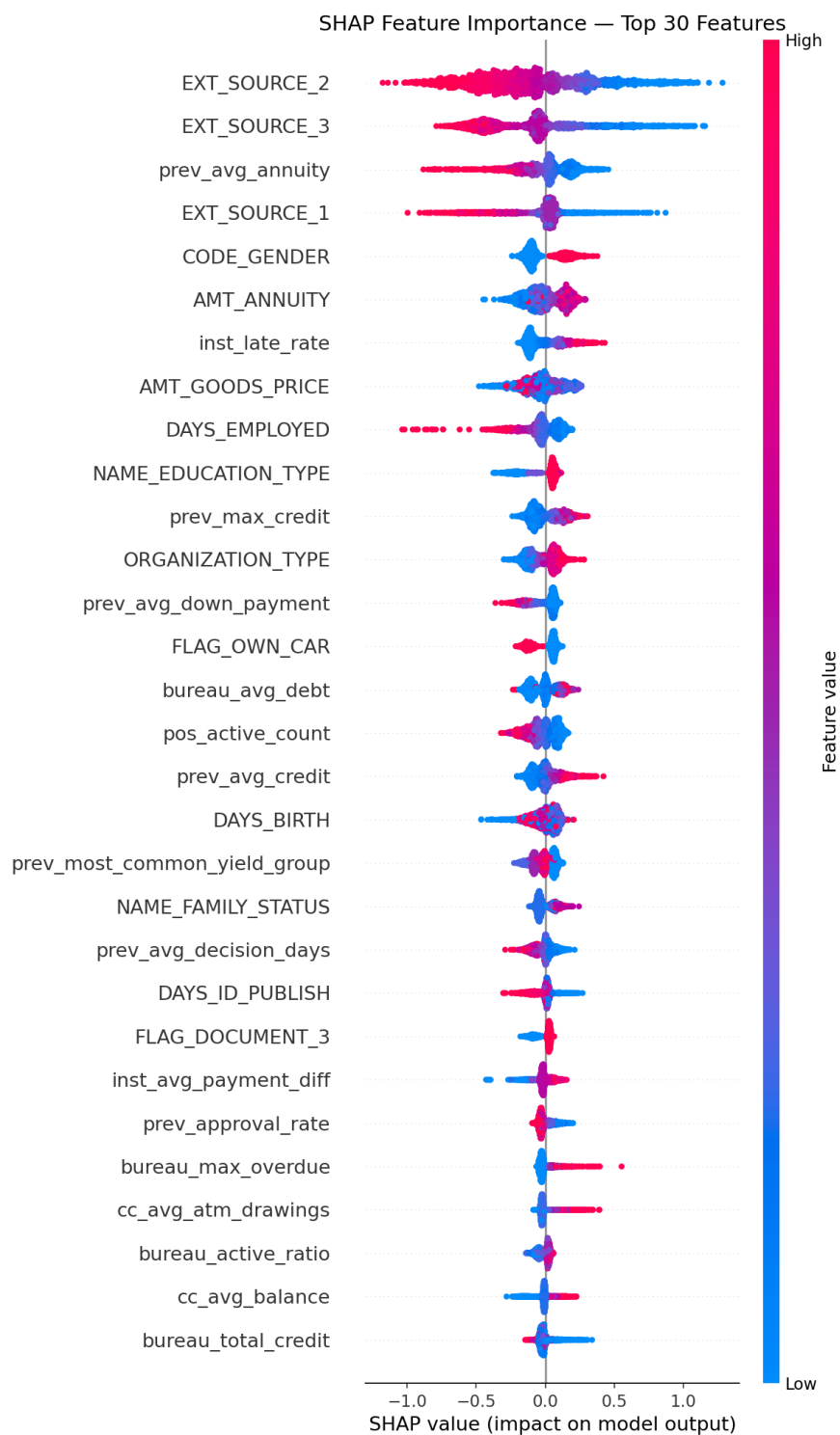


Figure 3. SHAP beeswarm plot — top 30 features by mean absolute SHAP value.

5.2 Mean Absolute SHAP Value

The bar plot shows the mean absolute SHAP value for each feature — a single summary of overall importance without directional information. Features with a mean absolute SHAP value near zero are candidates for removal in a second training run.

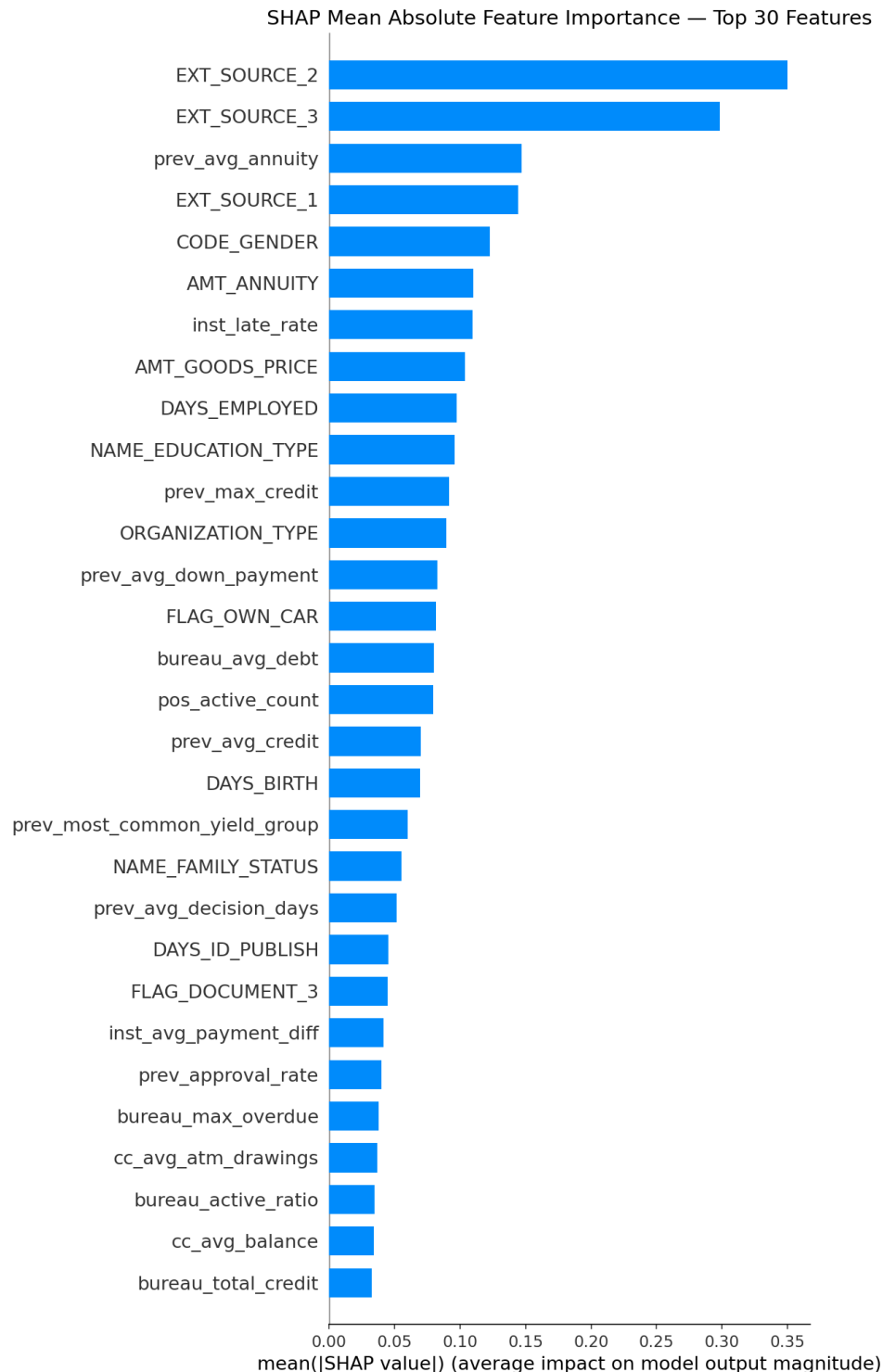


Figure 4. Mean absolute SHAP value per feature — top 30 features.